

CGPA calculation

Schema of table for given data:

```
create table student (  
    RegID int,  
    Ayear int,  
    Asem int,  
    PYS int,  
    CourseCode varchar(255),  
    CourseName varchar(255),  
    Credits int,  
    RegNo int,  
    Grade varchar(2),  
    Dept int  
);
```

Getting data from csv file

```
COPY student(RegID, Ayear, Asem, PYS, CourseCode, CourseName, Credits, RegNo, Grade, Dept)  
FROM 'C:/temp/Copy of Student-Data-2025 - Student-Data-2025.csv'  
DELIMITER ','  
CSV HEADER;
```

Creating more tables

```
create table registration(RegID int,RegNo int,Asem int,PYS int,CourseCode varchar(255)) ;  
create table course(CourseCode varchar(255),CourseName varchar(255),Credits int);  
create table gradepoint(grade varchar(2),points int);  
create table marksgrade(RegID int,grade varchar(2));
```

Inserting values into different tables created from given data.

```
insert into registration (regid,regno,asem,pys,coursecode)  
    (select regid,regno,asem,pys,coursecode from student);  
insert into course (coursecode,coursename,credits)  
    (select coursecode,coursename,credits from student);  
insert into marksgrade (regid,grade) SELECT regid,grade from student;
```

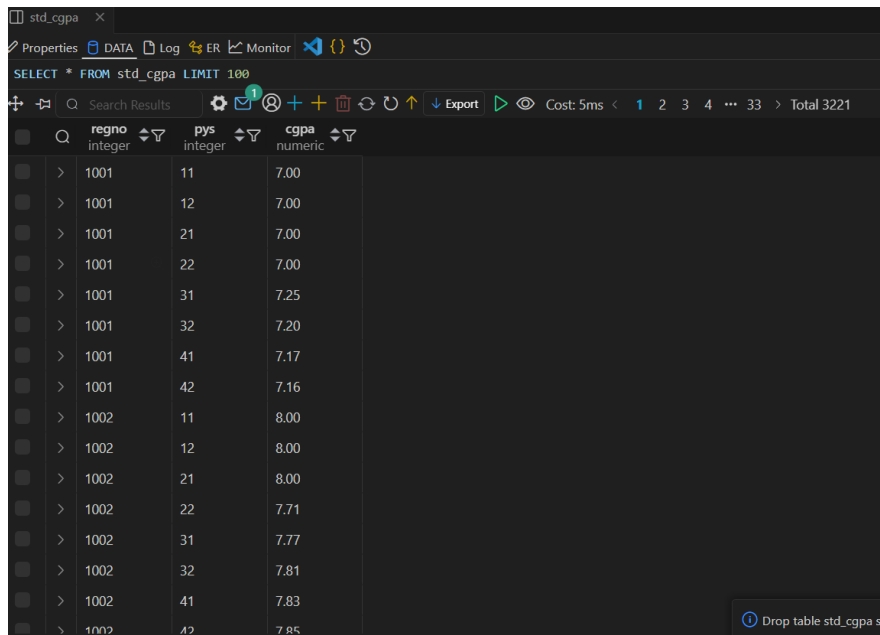
```
insert into gradepoint  
(grade,points)values('EX',10),('A',9),('B',8),('C',7),('D',6),('F',0),('P',5),('R',0),('X',0),('I',0);
```

creating gp_all table

```
create table gp_all as select r.regno,r.pys,r.coursecode,gp.grade,c.credits,gp.points from registration  
r join marksgrade gr on r.regid=gr.regid join course c on r.coursecode=c.coursecode join gradepoint  
gp on gr.grade=gp.grade group by r.regno,r.pys,gp.grade,r.coursecode,c.credits,gp.points order by  
r.regno,r.pys;
```


creating stds_cgpa table

create table stds_cgpa as select a.regno,a.pys,Round(sum(b.sgpa*b.sc)/sum(b.sc),2) as cgpa from std_sgpas a join std_sgpas b on a.regno=b.regno and a.pys>=b.pys GROUP BY a.regno,a.pys order by a.regno,a.pys;

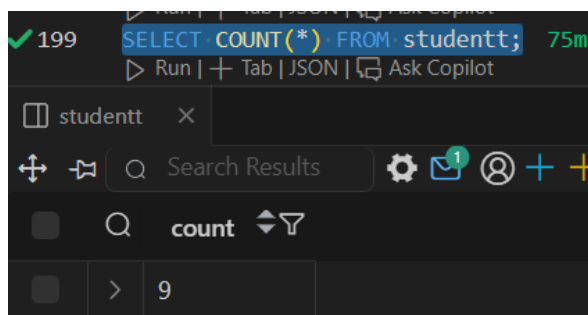


regno	pys	cgpa
1001	11	7.00
1001	12	7.00
1001	21	7.00
1001	22	7.00
1001	31	7.25
1001	32	7.20
1001	41	7.17
1001	42	7.16
1002	11	8.00
1002	12	8.00
1002	21	8.00
1002	22	7.71
1002	31	7.77
1002	32	7.81
1002	41	7.83
1002	42	7.85

LAB TASK

```
CREATE TABLE STUDENTt(  
    SID INT,  
    SNAME VARCHAR(255),  
    SBRANCH VARCHAR(255),  
    STOTAL FLOAT,  
    SCGPA NUMERIC(3,2),  
    DBMS FLOAT,  
    DAA FLOAT,  
    OOPS FLOAT,  
    COA FLOAT,  
    TOC FLOAT  
);
```

- a. Count the no.of students in the student table.



count
9

- b. Display the sum of DBMS marks of students whose marks are less than 17.5

```
200 SELECT SUM(DBMS) FROM STUDENTT WHERE DBMS<17.5 ; 2ms
```

Run

studentt x

Search Results

Cost: 4ms < 1 > Total 1

sum

10

- c. Display the count of students who failed DBMS(<17.5)

```
203 SELECT COUNT(SID) FROM STUDENTT WHERE DBMS<17.5; 3ms
```

204

studentt x

Search Results

Cost: 8ms < 1 > Total 1

count

1

- d. Display sid, sname of student who gets Maximum marks in DAA.

```
205 SELECT SID,SNAME FROM studentt WHERE DAA=(SELECT MAX(DAA) FROM STUDENTT); 2ms
```

Result(RO) x

Search Results

Cost: 2ms < 1

sid integer sname

423141 PARDHU

- e. Display Sid,sname,scgpa of students who gets minimum marks in OOPS

```
206 SELECT SID,SNAME,SCGPA FROM studentt WHERE OOPS=(SELECT MIN(OOPS) FROM STUDENTT); 3ms
```

207

Result(RO) x

Search Results

Cost: 3ms < 1 > Total 1

sid integer sname scgpa

623121 HRUTH 9.80

- f. Display Average marks in DBMS.

```
208 SELECT AVG(DBMS) FROM STUDENTT; 5ms
```

studentt x

Search Results

avg

26.25

- f. Display sid,sname,OOPS,DAA of students who failed in DAA and Passed in OOPS.

```
209 SELECT SID,SNAME,OOPS,DAA FROM STUDENTT WHERE DAA<17.5 AND OOPS>=17.5; 2ms
```

studentt

sid	sname	oops	daa
integer	varchar(255)	double precision	double precision

g. Display Minimum and maximum marks in All the 5 subjects.

```
355 select 'DBMS'as subject , max(dbms) as max_marks ,min(dbms) as min_marks from studentt
356 union
357 select 'DAA'as subject , max(daa) as max_marks ,min(daa) as min_marks from studentt
358 UNION
359 select 'COA'as subject , max(coa) as max_marks ,min(coa) as min_marks from studentt
360 UNION
361 select 'TOC'as subject , max(toc) as max_marks ,min(toc) as min_marks from studentt
362 UNION
363 select 'OOPS'as subject , max(oops) as max_marks ,min(oops) as min_marks from studentt; 3ms
364
```

Result(RO)

subject	max_marks	min_marks
COA	27	23
DAA	29	24
DBMS	45	10
OOPS	38	32
TOC	36	23

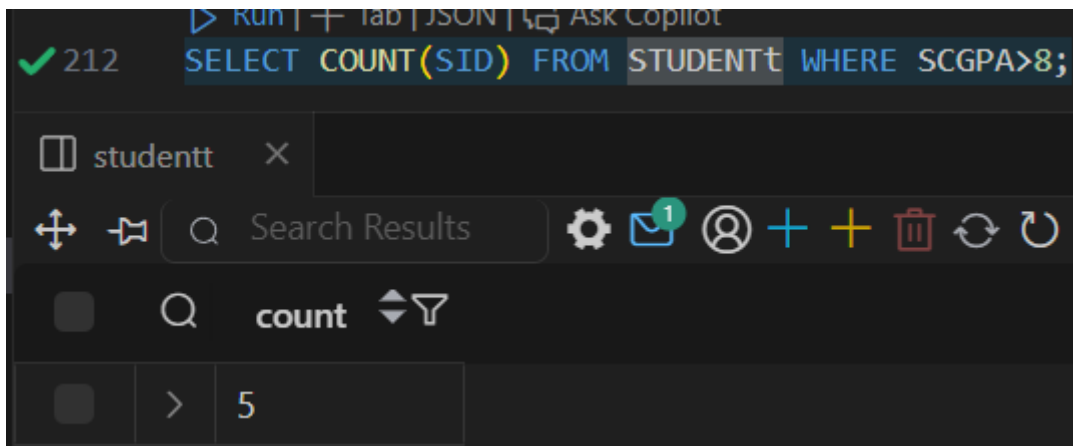
h. Display the count of Non CSE students in the table.

```
211 SELECT COUNT(*) FROM STUDENTT WHERE SBRANCH <>'CSE'; 5ms
```

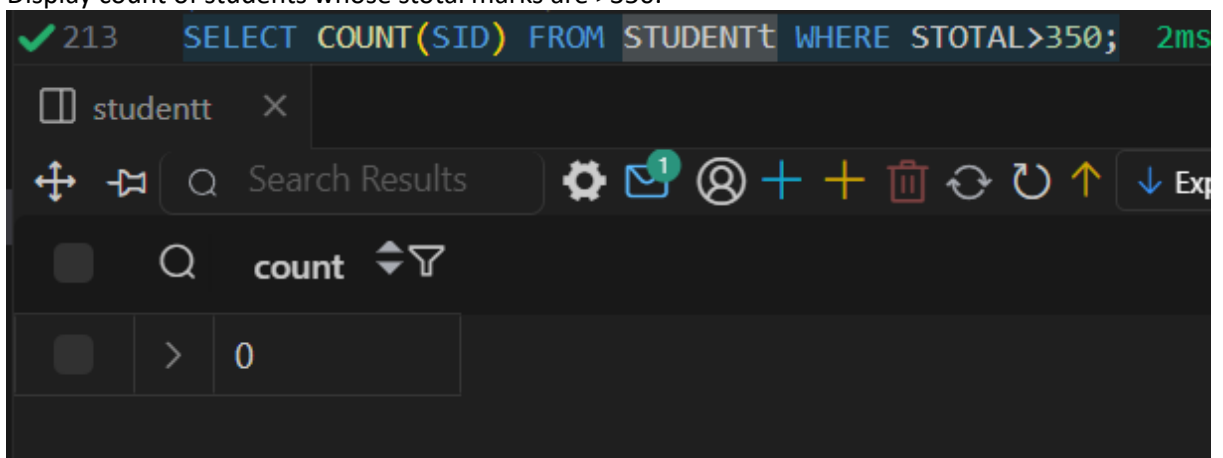
studentt

count
2

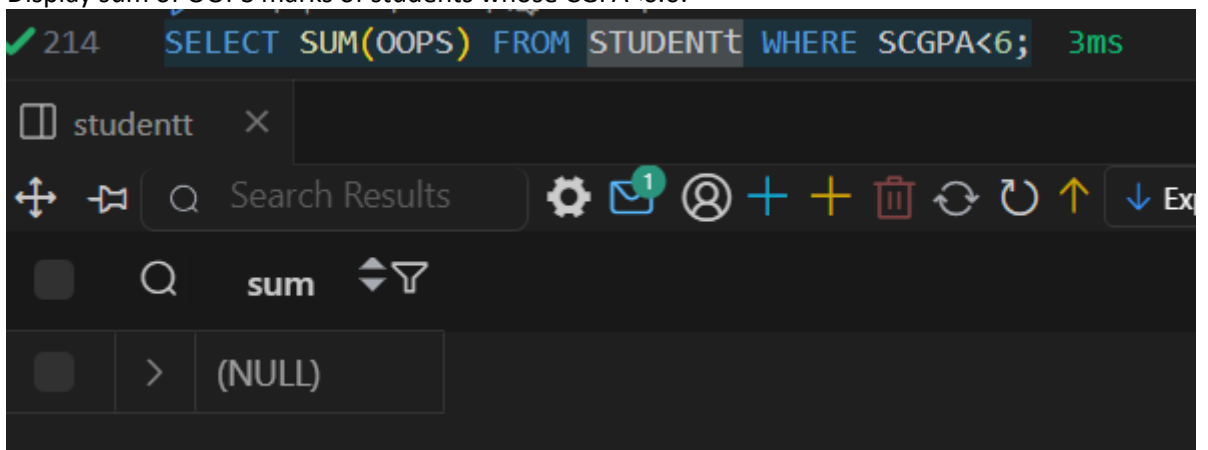
i. Display the count of students whose CGPA>8.0.



- j. Display count of students whose stotal marks are >350.

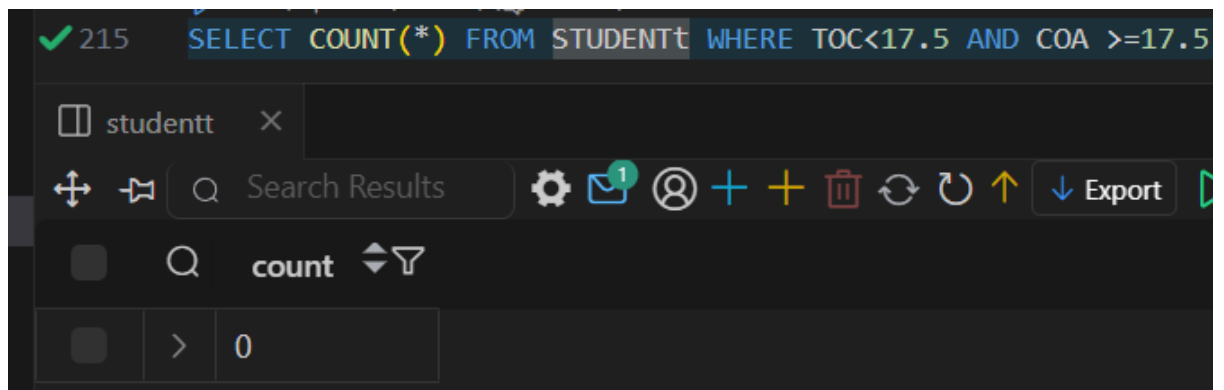


- k. Display sum of OOPS marks of students whose CGPA < 6.0.



- l. Display the count of students who failed in TOC and passed in COA.

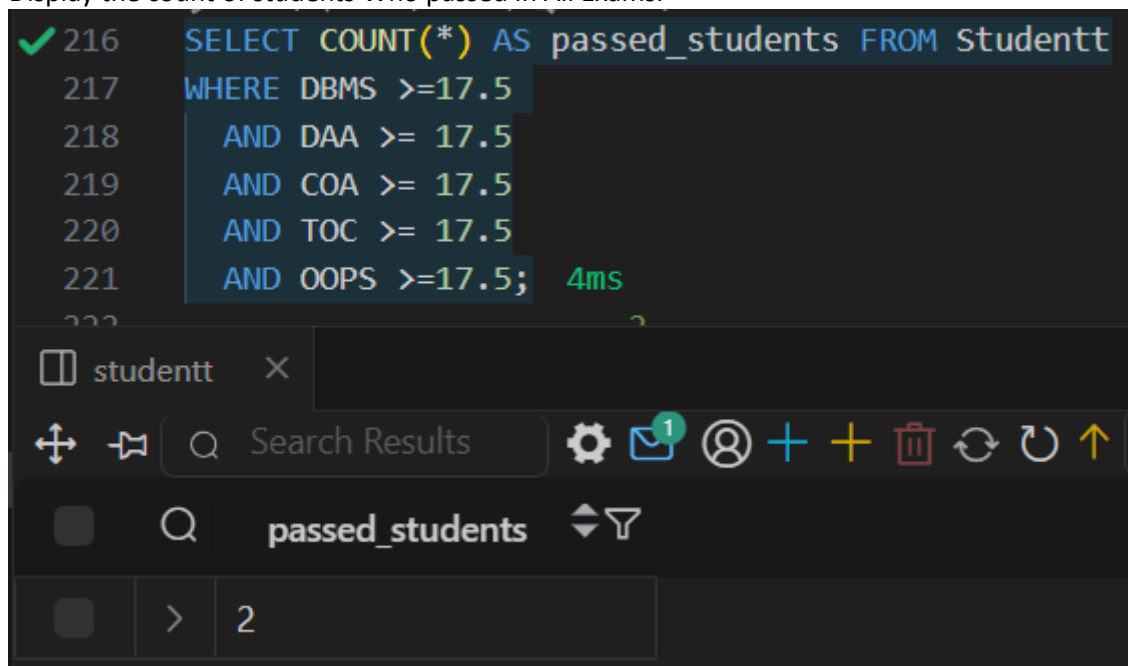
```
✓ 215 SELECT COUNT(*) FROM studentt WHERE TOC<17.5 AND COA >=17.5
```



The screenshot shows a SQL IDE interface. At the top, a query is entered: `SELECT COUNT(*) FROM studentt WHERE TOC<17.5 AND COA >=17.5`. Below the query editor, there is a toolbar with various icons for search, settings, and execution. The results pane shows a single row with the column name `count` and a value of `0`.

m. Display the count of students Who passed in All Exams.

```
✓ 216 SELECT COUNT(*) AS passed_students FROM Studentt
217 WHERE DBMS >=17.5
218 AND DAA >= 17.5
219 AND COA >= 17.5
220 AND TOC >= 17.5
221 AND OOPS >=17.5; 4ms
```



The screenshot shows a SQL IDE interface. The query editor contains the following SQL statement: `SELECT COUNT(*) AS passed_students FROM Studentt WHERE DBMS >=17.5 AND DAA >= 17.5 AND COA >= 17.5 AND TOC >= 17.5 AND OOPS >=17.5;`. The execution time is shown as `4ms`. Below the query editor, there is a toolbar with various icons for search, settings, and execution. The results pane shows a single row with the column name `passed_students` and a value of `2`.

2. Answer the following Questions

a. Display the Subject wise failed count of students

```

223 SELECT
224     'DBMS' AS subject, COUNT(*) AS failed_count
225 FROM Studentt
226 WHERE DBMS < 17.5
227 UNION ALL
228 SELECT
229     'DAA' AS subject, COUNT(*) AS failed_count
230 FROM Studentt
231 WHERE DAA < 17.5
232 UNION ALL
233 SELECT
234     'COA' AS subject, COUNT(*) AS failed_count
235 FROM Studentt
236 WHERE COA < 17.5
237 UNION ALL
238 SELECT
239     'TOC' AS subject, COUNT(*) AS failed_count
240 FROM Studentt
241 WHERE TOC < 17.5
242 UNION ALL
243 SELECT
244     'OOPS' AS subject, COUNT(*) AS failed_count
245 FROM Studentt
246 WHERE OOPS < 17.5; 2ms
247
248

```

Result(RO) ×

Search Results

Cost: 2ms < 1 > Total 5

subject failed_count
bigint

subject	failed_count
DBMS	1

b. Display Subject wise Maximum marks of all Subjects.

```

Run | + Tab | JSON | Ask Copilot
SELECT
    'DBMS' AS subject, MAX(DBMS) AS max_marks
FROM studentt
UNION ALL
SELECT
    'DAA' AS subject, MAX(DAA) AS max_marks
FROM studentt
UNION ALL
SELECT
    'COA' AS subject, MAX(COA) AS max_marks
FROM studentt
UNION ALL
SELECT
    'TOC' AS subject, MAX(TOC) AS max_marks
FROM studentt
UNION ALL
SELECT
    'OOPS' AS subject, MAX(OOPS) AS max_marks
FROM studentt; 4ms
Run | + Tab | JSON | Ask Copilot
SELECT

```

Result(RO) ×

Search Results

ms < 1 > Total 5

subject max_marks
double precision

subject	max_marks
DBMS	45

c. Display Subject wise Minimum marks of all Subjects.

```

✓ 268 SELECT
269     'DBMS' AS subject, MIN(DBMS) AS min_marks
270 FROM studentt
271 UNION ALL
272 SELECT
273     'DAA' AS subject, MIN(DAA) AS min_marks
274 FROM studentt
275 UNION ALL
276 SELECT
277     'COA' AS subject, MIN(COA) AS min_marks
278 FROM studentt
279 UNION ALL
280 SELECT
281     'TOC' AS subject, MIN(TOC) AS min_marks
282 FROM studentt
283 UNION ALL
284 SELECT
285     'OOPS' AS subject, MIN(OOPS) AS min_marks
286 FROM studentt; 3ms
287

```

Result(RO) X

Search Results

Cost: 3ms < 1 > Total 5

subject min_marks
double precision

subject	min_marks
DBMS	10

d. Display Branch wise student count

```

✓ 288 select sbranch , count(*) from studentt GROUP BY sbranch;
289

```

studentt X

Search Results

Cost: 3ms < 1 > Total 3

sbranch count
varchar(255)

sbranch	count
CSE	3

e. Display Subject wise Absents count .

```
✓ 299 SELECT
300     'DBMS' AS subject, COUNT(*) AS absent_count
301 FROM Studentt
302 WHERE DBMS IS NULL
303 UNION ALL
304 SELECT
305     'DAA' AS subject, COUNT(*) AS absent_count
306 FROM Studentt
307 WHERE DAA IS NULL
308 UNION ALL
309 SELECT
310     'COA' AS subject, COUNT(*) AS absent_count
311 FROM Studentt
312 WHERE COA IS NULL
313 UNION ALL
314 SELECT
315     'TOC' AS subject, COUNT(*) AS absent_count
316 FROM Studentt
317 WHERE TOC IS NULL
318 UNION ALL
319 SELECT
320     'OOPS' AS subject, COUNT(*) AS absent_count
321 FROM Studentt
```

Result(RO) X

Search Results

1

+

+

+

+

+

+

+

+

+

+

subject

absent_count

bigint

>	DBMS	1
>	DAA	0
>	COA	0
>	TOC	0
>	OOPS	1

f.Display Branch wise failure count

Run | Tab | JSON | Ask Copilot

```
290 select sbranch , count(*) from studentt WHERE dbms<17.5 or coa<17.5 or daa<17.5 or toc<17.5 or oops<17.5 GROUP BY sbranch; 2ms
```

studentt x

Search Results

Cost: 5ms < 1 > Total 1

sbranch	count
CSE	1

g. Display Branch wise Maximum and Minimum CGPA of Students

Run | Tab | JSON | Ask Copilot

```
291 select sbranch ,Max(scgpa) ,Min(scgpa) from studentt GROUP BY sbranch; 4ms
```

studentt x

Search Results

Cost: 6ms < > Total 1

sbranch	max	min
CSE	9.70	9.30
EEE	9.50	9.50

h. Display Branch wise Absentees count

Run | Tab | JSON | Ask Copilot

```
293 select sbranch ,count(*) from studentt where dbms is null or coa is null or daa is null or toc is null or oops is null GROUP BY sbranch;
```

studentt x

Search Results

Cost: 14ms < 1 > Total 2

sbranch	count
ECE	1
EEE	1

i. Display Subject wise sum of marks of students

Run | Tab | JSON | Ask Copilot

```
294 select sum(DBMS) as dbms,sum(DAA) as daa,sum(COA) as coa,sum(TOC) as toc,sum(OOPS) as oops from studentt; 2ms
```

studentt x

Search Results

Cost: 8ms < 1 > Total 1

dbms	daa	coa	toc	oops
105	129	123	173	138

j. Display Branch wise Sum of Marks of students

```
295 SELECT
296     sbranch,
297     SUM(COALESCE(DBMS, 0) +
298         COALESCE(COA, 0) +
299         COALESCE(TOC, 0) +
300         COALESCE(OOPS, 0) +
301         COALESCE(DAA, 0)) AS total_marks
302 FROM studentt
303 GROUP BY sbranch; 11ms
```

studentt x

Search Results

< 1 > Total 3

sbranch	total_marks
CSE	422
EEE	132
ECE	114

k. Display Subject wise Maximum marks for all Subjects whose maximum marks >35.

```
305 SELECT
306     'DBMS' AS subject, MAX(DBMS) AS max_marks
307 FROM studentt
308 HAVING MAX(DBMS) > 35
309 UNION
310 SELECT
311     'DAA' AS subject, MAX(DAA) AS max_marks
312 FROM studentt
313 HAVING MAX(DAA) > 35
314 UNION
315 SELECT
316     'COA' AS subject, MAX(COA) AS max_marks
317 FROM studentt
318 HAVING MAX(COA) > 35
319 UNION
320 SELECT
321     'TOC' AS subject, MAX(TOC) AS max_marks
322 FROM studentt
323 HAVING MAX(TOC) > 35
324 UNION
325 SELECT
326     'OOPS' AS subject, MAX(OOPS) AS max_marks
327 FROM studentt
328 HAVING MAX(OOPS) > 35; 11ms
```

Result(RO) X

Search Results

< 1 > Total 3

subject	max_marks
DBMS	45
OOPS	38

l. Display Subject wise Minimum marks for all Subjects whose minimum marks >17.5 and the marks are in descending order.

```
✓ 329 SELECT
330     'DBMS' AS subject, MIN(DBMS) AS min_marks
331 FROM studentt
332 HAVING MIN(DBMS) > 17.5
333 UNION
334 SELECT
335     'DAA' AS subject, MIN(DAA) AS min_marks
336 FROM studentt
337 HAVING MIN(DAA) > 17.5
338 UNION
339 SELECT
340     'COA' AS subject, MIN(COA) AS min_marks
341 FROM studentt
342 HAVING MIN(COA) > 17.5
343 UNION
344 SELECT
345     'TOC' AS subject, MIN(TOC) AS min_marks
346 FROM studentt
347 HAVING MIN(TOC) > 17.5
348 UNION
349 SELECT
350     'OOPS' AS subject, MIN(OOPS) AS min_marks
351 FROM studentt
352 HAVING MIN(OOPS) > 17.5
353 ORDER BY min_marks DESC; 3ms
354
```

Result(RO) ×

↕

🔍 Search Results

⚙️ 1 👤 + + 🗑️ ↺

< 1 > Total 4

🔍 subject 🔼🔽

min_marks
double precision 🔼🔽

>	TOC	33
>	OOPS	32

p. Display sid,same,cgpa in the descending order of scgpa.

```

✓ 406 SELECT 'DBMS' AS subject, SID FROM studentt WHERE DBMS = (
407     SELECT
408         MAX(DBMS)
409     FROM studentt
410 )
411 UNION SELECT 'DAA' AS subject, SID FROM studentt WHERE DAA = (
412     SELECT
413         MAX(DAA)
414     FROM studentt
415 )
416 UNION SELECT 'COA' AS subject, SID FROM studentt WHERE COA = (
417     SELECT
418         MAX(COA)
419     FROM studentt
420 )
421 UNION SELECT 'TOC' AS subject, SID FROM studentt WHERE TOC = (
422     SELECT
423         MAX(TOC)
424     FROM studentt
425 )
426 UNION SELECT 'OOPS' AS subject, SID FROM studentt WHERE OOPS = (
427     SELECT
428         MAX(OOPS)
429     FROM studentt
430 ) 2ms

```

Result(RO) X
















Cost:

< 1 > Total 7

integer

>	COA	423141
>	COA	523141
>	COA	623141

✓ 404 `select sid,sname,scgpa from studentt ORDER BY scgpa DESC;` 2ms

		sid integer	sname varchar(255)	scgpa numeric(3,2)
☐	>	623121	HRUTH	9.80
☐	>	423141	PARDHU	9.70
☐	>	523141	AJAY	9.50
☐	>	423513	RANDOM	9.30
☐	>	423113	SAKETH	9.30

Display Sid,sname,stotal,scgpa in the descending order of stotal

✓ 402 `select sid,sname,stotal,scgpa from studentt ORDER BY stotal DESC;`

		sid integer	sname varchar(255)	stotal double precision	scgpa numeric(3,2)
☐	>	423141	PARDHU	155	9.70
☐	>	423513	RANDOM	141	9.30
☐	>	423113	SAKETH	141	9.30
☐	>	523141	AJAY	132	9.50
☐	>	623121	HRUTH	114	9.80

Display branch wise Sid who got maximum CGPA

```
✓ 391 SELECT
    392     sbranch,
    393     SID
    394 FROM studentt s
    395 WHERE SCGPA = (
    396     SELECT
    397         MAX(SCGPA)
    398     FROM studentt
    399     WHERE sbranch = s.sbranch
    400 ) 2ms
```

Result(RO) X

Search Results

Q

sbranch

sid

integer

>

CSE

423141

>

ECE

623121

>

EEE

523141